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Insup Lee

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Research Interests

My research lies at the intersection of security and AI systems. I currently work on accelerating robot demonstrations for faster task execution with VLA policies, and aim to improve the safety and trustworthiness of physical AI.

Education

Korea University — Seoul, Korea
Ph.D. in Cybersecurity (Advisors: Sangjin Lee and Seokhie Hong) — Sep 2019 - Aug 2026

Korea University — Seoul, Korea
B.E. in Cyber Defense — Mar 2014 - Feb 2018

Work Experience

KAIST AI — Seoul, Korea
Postdoctoral Researcher, LLM Innovation Research Center (Advisor: Dongsu Han) — Sep 2026 - Present

Ministry of National Defense — Seoul, Korea
Cybersecurity Officer (Army Captain), Republic of Korea Army — Aug 2023 - May 2025

Agency for Defense Development (ADD) — Seoul, Korea
Researcher, Cyber Technology Center — Jul 2018 - Jul 2023

Publications

(C: conference, J: journal, P: preprint, †: corresponding author)

International Publications

C5 *Schema-Agnostic Knowledge Graph Construction via Hybrid Ontology Discovery for Cyber Threat Intelligence*
S. Kim, J. Kim, D. Kang, D. Kim, and **I. Lee**†
Annual Computer Security Applications Conference (**ACSAC'26**), Dec. 2026.

C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, D. Bae, G. Kim, G. Ahn, **I. Lee**, S. Park, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (**CCS'26**), Nov. 2026.

J11 *MoDiT: Domain-Specific Diffusion Transformers for Automatic Modulation Classification*
I. Lee† and M. Alghfeli
IEEE Transactions on Cognitive Communications and Networking (**TCCN**), 2026.

J10 *Multi-Domain Side-Channel Analysis for Anomaly Detection in Embedded Systems*
S. Park, D. Bae, **I. Lee**, J. Kim, H. Oh, S. Hong, and H. Kim
IEEE Embedded Systems Letters, 2026.

C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (**DAC'26**), Jul. 2026.

J9 *MuCamp2: Generating Validated Cyber Campaign Variants via Constrained LLMs for Group Attribution*
K. Kim, J. Lee, **I. Lee**, I. Han, J. Lee, S. Shim, and C. Choi
IEEE Access, Vol. 14, pp. 87329-87344, Jun. 2026.

J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*
I. Lee, D. Bae, S. Hong, and S. Lee
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.

J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.

J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*
I. Lee†, K. Alteneiji, and M. Alghfeli
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.

J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*
I. Lee and C. Choi
IEEE Transactions on Information Forensics and Security (**TIFS**), Vol. 20, pp. 6162-6174, Jun. 2025.

J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*
I. Lee and W. Lee
IEEE Transactions on Dependable and Secure Computing (**TDSC**), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.

J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*
I. Lee and C. Choi
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.

J2 *Exploiting TTP Co-occurence via GloVe-Based Embedding With ATT&CK Framework*
C. Shin, **I. Lee**, and C. Choi
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.

J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.

C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*
S. Shin, **I. Lee**, and C. Choi
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.

C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*
C. Choi, S. Shin, and **I. Lee**
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

Domestic Publications (Korean)

D4 S. Park, W. Shin, D. Bae, H. Shin, **I. Lee**, S. Hong, and H. Kim, "Single Fault Injection and Countermeasure for MAVLink Authentication Enforcement in Unmanned Aerial Vehicles," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2026. (Selected as an Outstanding Paper Award)

D3 S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)

D2 H. Park and **I. Lee**†, "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.

D1 K. Kim and **I. Lee**†, "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

Other Experience

AI Cyber Challenge (AIxCC), DARPA and ARPA-H, USA — Apr 2024 – Aug 2024

- Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities

SW Outsourcing Development, KCMVP-Certified Cryptographic Module — Jun 2017 – May 2018

- Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)

Honors and Awards

- Outstanding Paper Award — Nov 2026
  CISC-W'26, KIISC
- KU Graduate School Achievement Award — Feb 2026
  Korea University, Seoul, Republic of Korea
- Outstanding Paper Award — Nov 2025
  CISC-W'25, KIISC
- Certificate of Commendation (UAE-ROK Engagement Program) — Mar 2025
  United Arab Emirates Ministry of Defense
- Ambassador's Commendation — Mar 2025
  Embassy of the Republic of Korea to the United Arab Emirates
- Full Tuition Scholarship (Korea University) — Mar 2014 – Feb 2018
  Ministry of National Defense, Republic of Korea

Academic Service

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE/ACM Transactions on Networking (ToN), 2026
- IEEE Transactions on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

Teaching Experience

- Lecturer, Computer Networks (SCS302, Fall 2025), Korea University — Sep 2025 - Feb 2026
- Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense — Sep 2024
- Instructor, Penetration Testing - Basic, UAE Ministry of Defense — Jul 2024

Mentoring Experience

- **Seonwoo Kim** (Major at Ministry of National Defense) — Feb 2026 – Present
  Topic: LLM-based knowledge graph construction for CTI [C5]
- **Sujin Park** (Ph.D. Student at Korea University) — Jun 2025 – Present
  Topic: Side-channel analysis for anomaly detection [J10], [D3], [D4]
- **Hyoungrok Kim** (M.S. Student at Korea University - currently at XBOW) — Mar 2025 – Sep 2025
  Topic: LLM-based TTP extraction for CTI [J7]
- **Hyunjun Park** (Navy Lieutenant at Ministry of National Defense) — Nov 2024 – Feb 2025
  Topic: DDoS detection via transfer learning [D2]
- **Kangmun Kim** (First Lieutenant at Ministry of National Defense) — Jan 2024 – Sep 2024
  Topic: Web shell detection via user behavior embedding [D1]

Reference

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Professor Emeritus, School of Cybersecurity, Korea University
Former Special Presidential Advisor for Cybersecurity, Republic of Korea
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